

Simulation report - Experiment 2

Streaming CA-CFAR verification evidence

DUTs: cfar_core and radar_demo. Original SystemVerilog testbenches and exact source bytes match the successful Vivado build. Supplied local simulation: Icarus Verilog 12.0, 8 October 2026. Independent laptop confirmation: Vivado XSim 2025.1.1, same day.

Test suite	Observed pass output
tb_cfar_core	CORE SIMULATION PASS: checked=10721 accepted=12963
tb_radar_demo	DEMO SIMULATION PASS: 33 frame checks

Reference model: independent chronological sample array and expected-result queue. It checks full-width arithmetic, valid timing, index alignment and detection against each emitted DUT result. Coverage includes equality, training/guard exclusion, integer truncation, unsigned high values, display saturation, continuous/gapped traffic, random values/scales, short frames and reset.

Controller: four scenarios, four factors and two gap modes produce 32 combinations; one repeated standard frame gives 33 checks. The testbench additionally checks held-start behavior and reset during processing.

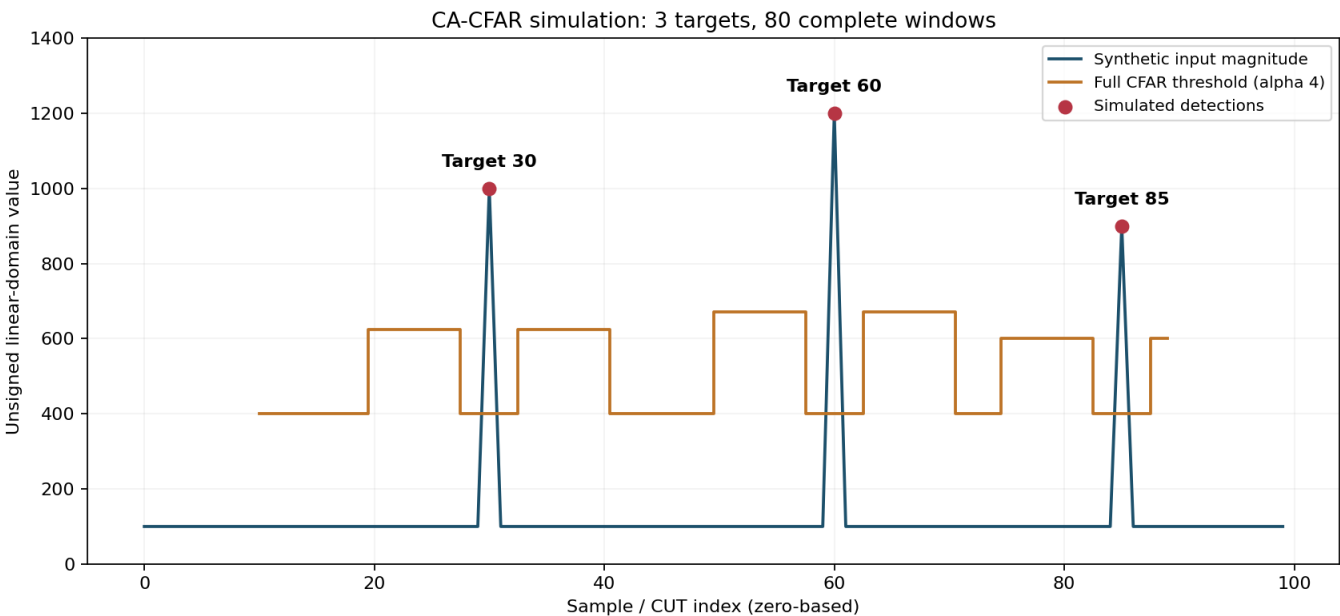


Figure S1. Plot from the original standard-frame result CSV. Exactly 80 full windows and three detections at 30, 60, 85; alpha four. Magnitudes are controlled synthetic unsigned linear-domain values, not RF measurements.

Waveform, timing and expected results

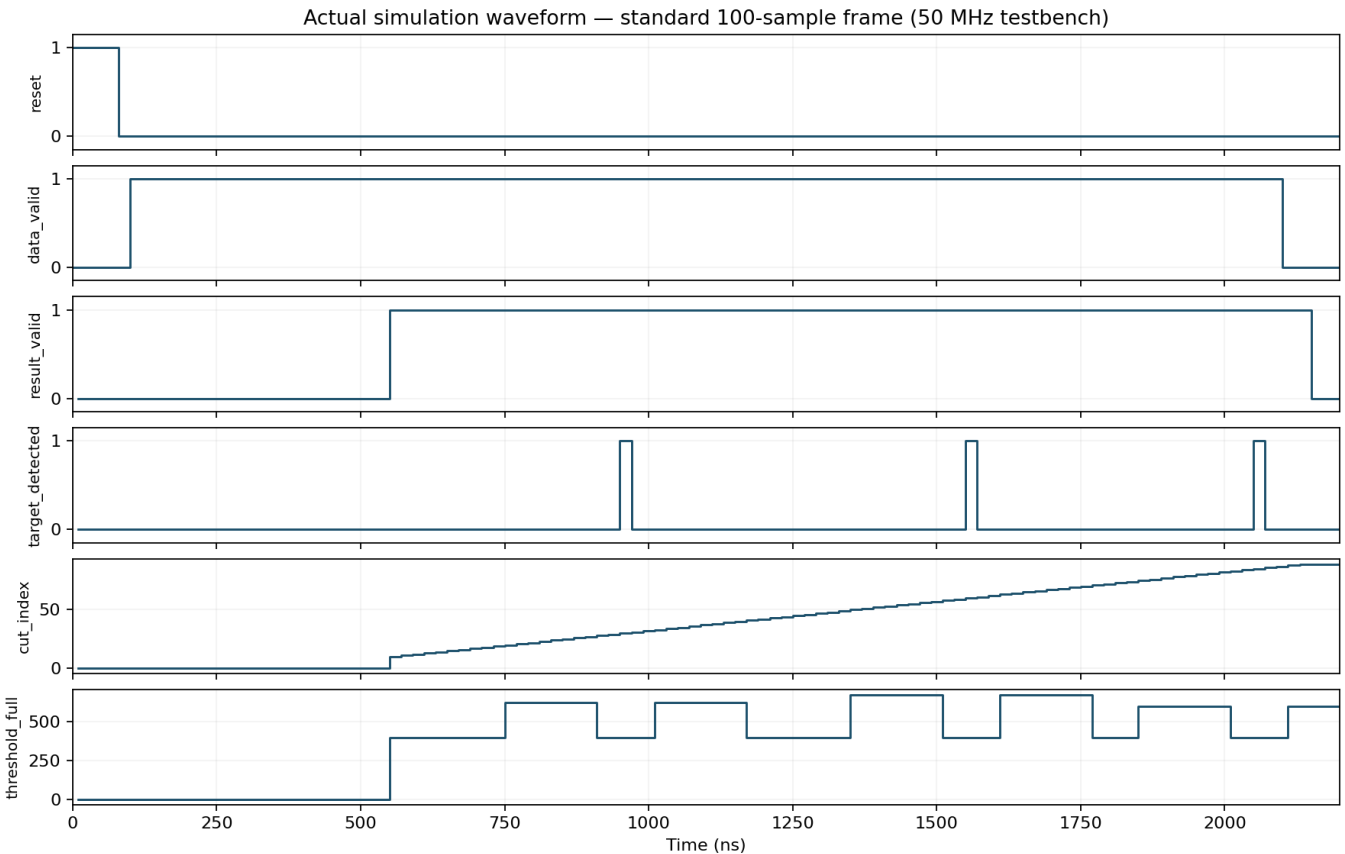


Figure S2. Real supplied cfar_core.vcd rendered as traces. Reset clears pending results; result_valid starts only after a full window and pipeline latency. Threshold_full is the comparison value, distinct from a saturating display. Full original traces are retained in Simulation/.

Case at alpha four	Detection result	Frame results
Constant 100	0 targets	80
1000/1200/900 at 30/60/85	3 targets: 30, 60, 85	80
Weak 300 at 30	0; factor two produces 1 at 30	80
20000 background / 60000 target	0: compare against full 80000 threshold	80

A result for CUT n-10 appears two clocks after the accepted newest sample n (n at least 20). Stalls change wall-clock spacing but never duplicate a sample window. The controller captures/drains outputs and finishes in 104 continuous cycles or 302 gapped cycles. At the confirmed 50 MHz design clock these equal 2.08 and 6.04 microseconds; host/JTAG time is excluded.

Transcripts and reproduction

The complete exact local output is Simulation/transcript.txt. The full successful XSim/synthesis/routing run is Evidence/vivado_build.log. Representative unmodified local transcript lines follow:

```
CORE SIMULATION PASS: checked=10721 accepted=12963
DEMO PASS sc=1 alpha=4 gap=0 detections=3 results=80 cycles=104
DEMO PASS sc=1 alpha=4 gap=1 detections=3 results=80 cycles=302
DEMO PASS sc=2 alpha=4 gap=0 detections=0 results=80 cycles=104
DEMO PASS sc=2 alpha=2 gap=0 detections=1 results=80 cycles=104
DEMO SIMULATION PASS: 33 frame checks
```

Reproduce locally with Icarus

Run from the experiment root with iverilog/vvp installed. The testbenches save VCD/CSV/PASS outputs. Use a separate copy to preserve this original archive.

```
mkdir -p Results Simulation
iverilog -g2012 -s tb_cfar_core -o core.vvp \
  RTL/cfar_core.v Testbench/tb_cfar_core.sv
vvp core.vvp +OUTDIR=.
iverilog -g2012 -s tb_radar_demo -o demo.vvp \
  RTL/cfar_core.v RTL/radar_demo.v Testbench/tb_radar_demo.sv
vvp demo.vvp +OUTDIR=.
```

For the complete vendor flow, RUN_PROJECT2.cmd invokes Vivado/build_project.tcl, which runs both XSim suites and requires their PASS markers before synthesis/routing. This flow recreates VIO and the native project. The recovered original native source archive is also included in FPGA_Project/. A new build clears old markers and hardware outputs; preserve the original archive first.

Schematic and evidence audit

The Yosys structural check generates the actual top schematic. Simulation/vio_elaboration_stub.v and vendor clock-library declarations serve as black boxes; this check does not verify their implementation, lock, routing or timing. Images contains the DOT/PNG/SVG; Simulation contains the used-module JSON and complete tool transcript. The generation recipe is Tools/SCHEMATIC_REPRODUCTION.md.

Run python Tools/validate_evidence.py for source/bitstream identity and saved CSV consistency. PDF generation itself does not execute hardware or simulation. A separate 10 October XSim rerun passed both suites; its logs are retained in Validation_2026-10-10/. No new board run was performed. Old September standalone core or AXI overlay results are not used as evidence for this complete October design.